

GREAT WALL STATION, Antarctica

WMO: 890580

Lat: 62.217S

Lon: 58.967W

Elev: 10

StdP: 101.20

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-16.6	-14.6	-19.1	0.7	-16.4	-16.9	0.9	-14.0	27.0	-5.6	24.9	-5.6	7.9	110	1.155

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.3	4.6	3.3	4.0	2.9	3.6	2.6	3.8	4.1	3.3	3.6	3.0	3.3	7.9	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
3.4	4.8	3.8	3.0	4.7	3.4	2.5	4.5	3.1	16.6	4.3	15.7	3.8	15.1	3.4	6.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.7	21.4	18.1	-17.7	6.6	2.7	1.6	-19.7	7.8	-21.3	8.7	-22.9	9.6	-24.9	10.7		
(5)			-17.8	4.5	2.8	0.6	-19.8	5.0	-21.4	5.3	-22.9	5.7	-24.9	6.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.0	1.7	1.8	0.6	-1.5	-2.8	-5.0	-5.8	-5.1	-4.3	-2.6	-0.9	0.4		
	DBStd	3.83	1.04	1.39	2.01	3.02	2.94	4.00	4.13	3.74	3.47	2.58	1.56	1.33		
	HDD10.0	4372	257	229	292	344	397	452	489	469	428	392	326	297		
	HDD18.3	7414	516	463	551	594	655	702	747	728	678	651	576	555		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	8.3	7.0	7.5	7.7	8.4	8.2	8.6	8.4	9.5	9.1	9.4	7.8	7.7		
(15) Precipitation	PrecAvg	684	59	66	72	63	62	53	54	62	58	55	45	43		
	PrecMax	821	126	97	123	135	105	134	140	160	105	116	82	79		
	PrecMin	471	20	12	10	32	27	6	12	26	27	23	12	14		
	PrecStd	103	25	18	26	24	21	26	25	30	20	22	15	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	5.6	5.0	3.7	2.4	1.6	0.8	0.9	1.4	2.2	3.3	5.2		
		MCWB	3.7	4.1	4.0	3.0	1.6	1.0	0.3	0.3	0.7	1.3	2.0	3.0		
	2%	DB	4.2	4.5	4.1	2.9	1.8	0.9	0.3	0.2	0.6	1.1	2.1	3.7		
		MCWB	2.9	3.5	3.3	2.4	1.2	0.4	-0.1	-0.2	0.1	0.5	1.2	2.1		
	5%	DB	3.7	4.0	3.5	2.4	1.3	0.4	-0.1	-0.1	0.2	0.7	1.6	3.0		
		MCWB	2.5	3.1	2.9	1.9	0.9	0.0	-0.5	-0.4	-0.2	0.1	0.8	1.7		
	10%	DB	3.2	3.5	3.1	1.8	0.9	-0.1	-0.6	-0.5	-0.2	0.3	1.1	2.4		
		MCWB	2.2	2.7	2.6	1.4	0.5	-0.4	-0.9	-0.8	-0.5	-0.1	0.5	1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.7	4.3	3.2	2.0	1.3	0.6	0.5	1.0	1.5	2.2	3.5		
		MCDB	4.6	5.2	4.8	3.4	2.1	1.4	0.7	0.5	1.2	1.9	2.9	4.3		
	2%	WB	3.4	3.9	3.6	2.6	1.5	0.6	0.1	0.0	0.2	0.7	1.6	2.6		
		MCDB	3.8	4.2	3.9	2.8	1.6	0.8	0.2	0.1	0.4	0.9	1.9	3.3		
	5%	WB	2.9	3.4	3.1	2.1	1.0	0.1	-0.4	-0.4	0.0	0.3	1.1	2.0		
		MCDB	3.4	3.7	3.4	2.3	1.3	0.3	-0.2	-0.2	0.1	0.5	1.4	2.6		
	10%	WB	2.5	3.0	2.7	1.5	0.6	-0.4	-0.9	-0.7	-0.4	0.0	0.7	1.5		
		MCDB	3.0	3.3	3.0	1.8	0.8	-0.2	-0.6	-0.5	-0.2	0.2	1.0	2.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.4	2.3	2.5	2.5	3.0	3.4	4.2	3.7	3.7	2.8	2.5	2.5		
		MCDBR	3.0	3.0	2.5	2.4	2.4	2.7	3.3	3.0	3.1	2.6	2.9	3.7		
	5% WB	MCWBR	2.3	2.4	2.3	2.4	2.7	2.8	3.5	3.1	3.0	2.5	2.4	2.7		
		MCWBR	2.7	2.6	2.4	2.3	2.4	2.7	3.5	2.9	3.0	2.5	2.5	3.2		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.308	0.297	0.280	0.225	0.366	0.211	0.269	0.288	0.310	0.304	0.309		
		taud	2.494	2.503	2.503	2.373	2.099	2.820	2.048	2.239	2.370	2.402	2.458	2.467		
	Ebn at Noon		967	918	841	675	442	301	437	660	828	903	966	978		
		Edh at Noon	98	87	71	53	31	20	31	57	78	95	101	104		
(44) All-Sky Solar Radiation	RadAvg		5.03	3.63	1.95	0.85	0.25	0.10	0.18	0.63	1.67	3.10	4.75	5.70		
	RadStd		0.46	0.28	0.16	0.07	0.02	0.00	0.01	0.05	0.13	0.21	0.29	0.45		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	-0.26	-0.42	N/A	N/A	N/A	N/A

Nomenclature: See separate page